Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
C		2613942

Dancing with Data: A Mathematical Framework for Fan Vote Estimation and Voting System Optimization in DWTS

Summary

Dancing with the Stars balances technical expertise with audience popularity, yet anomalies like Bobby Bones Season 27 victory expose flaws in this hybrid mechanism. The core challenge is the **information asymmetry** caused by confidential fan votes. We develop a framework to **reconstruct** hidden votes, **diagnose** systemic failures, and **engineer** a fairer **Dynamic Weighted Voting System (DWVS)**.

For Problem 1, we treat fan vote estimation as an inverse problem, solved via a **Constrained Optimization Model** using a Softmax function. Stratified analysis validates our model, achieving **80.7%** consistency with historical eliminations (peaking at **96.0%** in percentage-based seasons). We address uncertainty via **Bootstrap resampling**, yielding a **Certainty Index of 0.995**, proving our estimates are statistically robust.

For Problem 2, using these estimates, **Counterfactual Analysis** reveals that rank-based and percentage-based methods yield identical outcomes in **73.1%** of cases. However, percentage scoring disproportionately favors "high-popularity, low-skill" contestants. For controversial figures like Rice, Cyrus, Palin, and Bones, switching methods alters results in only 25% of cases. Crucially, the **Judges Tiebreaker Rule** is the primary defense, changing **29.7%** of decisions and effectively blocking mid-tier outliers like Bristol Palin.

For Problem 3, using **Analysis of Variance (ANOVA)** and **Random Forest**, we find external factors like age or industry have limited predictive power ($R^2 = 0.11$). Controversies stem not from demographic bias but a **static weight balance** that lets fanbases override judges in finals.

For Problem 4, we propose a Dynamic Weighted Voting System (DWVS) optimized via grid search: base α=0.35, increment=0.03, yielding initial fan weight of 65% shifting to 35% by finals. Simulation across all 32 identified controversial contestants shows 78.1% would rank lower under DWVS, with average adjustment of +0.34 positions. Bobby Bones would place 2nd instead of 1st.

In conclusion, our framework provides a scientifically grounded solution that restores the critical balance between democratic participation and professional integrity.

Keywords: Fan Vote Estimation; Constrained Optimization; Counterfactual Analysis; Dynamic Weighted Voting System; Bootstrap Resampling

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1 Introduction

1.1 Problem Background: The Bobby Bones Controversy

In Season 27 of Dancing with the Stars (DWTS), country radio host Bobby Bones achieved an outcome that sparked widespread debate: he won the Mirrorball Trophy despite ranking **7th-10th among judges throughout the competition**—consistently in the bottom third of technical scores. This was not an isolated incident. Jerry Rice reached the finals in Season 2 despite poor judge scores; Bristol Palin spent 50% of her weeks in elimination position yet placed 3rd in Season 11.

These controversial outcomes expose a fundamental tension in DWTS's hybrid voting system: **when does fan enthusiasm cross the line from democratic participation to unfair manipulation of results?** This research takes these controversies as our entry point, developing mathematical models to (1) quantify the hidden fan vote distribution, (2) understand what enables such "upset victories," and (3) design a fairer voting system that preserves entertainment value while ensuring technical merit is appropriately rewarded.

DWTS combines professional judge scores with audience votes to determine weekly eliminations. Since 2005, the show has completed 34 seasons with varying voting rules: rank-based combination in Seasons 1-2 and 28-34, percentage-based combination in Seasons 3-27, and a judge tiebreaker rule starting in Season 28. The exact fan vote totals have never been disclosed, creating information asymmetry that complicates fairness analysis.

1.2 Problem Restatement

We address four interconnected research questions forming a logical progression: (1) estimate fan votes with consistency and certainty measures—essential for understanding controversial outcomes; (2) compare rank-based and percentage-based voting methods—to determine if method choice enables controversies; (3) analyze the impact of professional dancers and celebrity characteristics—to identify systematic biases; and (4) design an improved voting system—to resolve the fairness issues identified in Problems 1-3.

1.3 Our Approach: From Controversy to Solution

Our research follows a "**Controversy** $\rightarrow$ **Modeling** $\rightarrow$ **Optimization**" framework. We first develop a constrained optimization model to estimate hidden fan votes (Problem 1), revealing that 32 controversial contestants systematically outperformed their judge rankings. We then apply counterfactual analysis (Problem 2) and factor analysis (Problem 3) to understand the mechanisms enabling such outcomes. Finally, we propose a **Dynamic Weighted Voting System (DWVS)** (Problem 4) that addresses fairness concerns while maintaining audience engagement. Cross-season validation (R^2=0.934 on held-out seasons) ensures our conclusions generalize to future competitions.

2 Preparation for Modeling

2.1 Model Assumptions

- **Assumption 1:** Fan votes are positively correlated with a contestant's underlying "popularity," which partially relates to their judge scores but includes additional factors not captured in the data.
  Justification: Contestants with better performances generally receive more fan support, though exceptions exist for highly popular celebrities with dedicated fan bases.
- **Assumption 2:** The voting rules announced by DWTS accurately reflect the actual elimination mechanism.
  Justification: As a major television production, DWTS has strong incentives to maintain credibility by following stated rules.
- **Assumption 3:** Judge scores accurately reflect relative dancing ability on a given week, with quantifiable inter-judge variance.
  Justification: Professional judges apply consistent standards. Our analysis reveals inter-judge score standard deviation averages 0.65 points, indicating acceptable consistency.

2.2 Notations

Table 1 summarizes the key mathematical symbols used throughout this paper.

Table 1: Summary of Key Notations

Symbol	Description	Symbol	Description
s_i	Judge score for contestant i	v_i	Estimated fan vote share
$\bar{s}$	Mean judge score	α_i	Popularity factor
n	Number of contestants	w	Week number
ϵ	Gaussian noise	B	Bootstrap iterations
σ_v, μ_v	Std. dev. and mean of v	R_i^{judge}, R_i^{fan}	Judge/fan rankings
P_i^{judge}, P_i^{fan}	Judge/fan percentages	$S_i(w)$	Comprehensive score
J_i	Judge score % (normalized)	F_i	Fan vote % (normalized)
$\alpha(w)$	Dynamic judge weight	T_i	Threshold penalty
F	ANOVA F-statistic	r	Correlation coefficient
CV	Coefficient of variation	R^2	Determination coefficient

2.3 Data Overview and Supplementary Collection

The primary dataset (COMAP) contains 421 contestants across 34 seasons with weekly judge scores. We supplemented this with Wikipedia pageviews (183 contestants, S21–34) via Wikimedia API[1] to capture baseline popularity, and aggregated pro dancer statistics (60 dancers) from the provided data. Season metadata was verified against official sources[2].

[1] `https://wikimedia.org/api/rest_v1/`
[2] Verified against Wikipedia DWTS pages and Disney+ official site.

Key Insight: Bobby Bones (S27 winner) ranked only 8th/13 in pageviews (273,153 vs. season avg. 318,942), suggesting his victory stemmed from fan mobilization efficiency rather than pre-existing fame.

Data Preprocessing: We transformed the wide-format data into 2,777 contestant-week observations with engineered features: weekly rankings (R_i^{judge}, P_i^{judge}), cumulative performance (rolling averages), elimination flags, and voting method indicators. Missing 4th-judge scores (65–94% missing) were retained as they result from structural elimination patterns.

2.4 Our Work: Methodological Framework

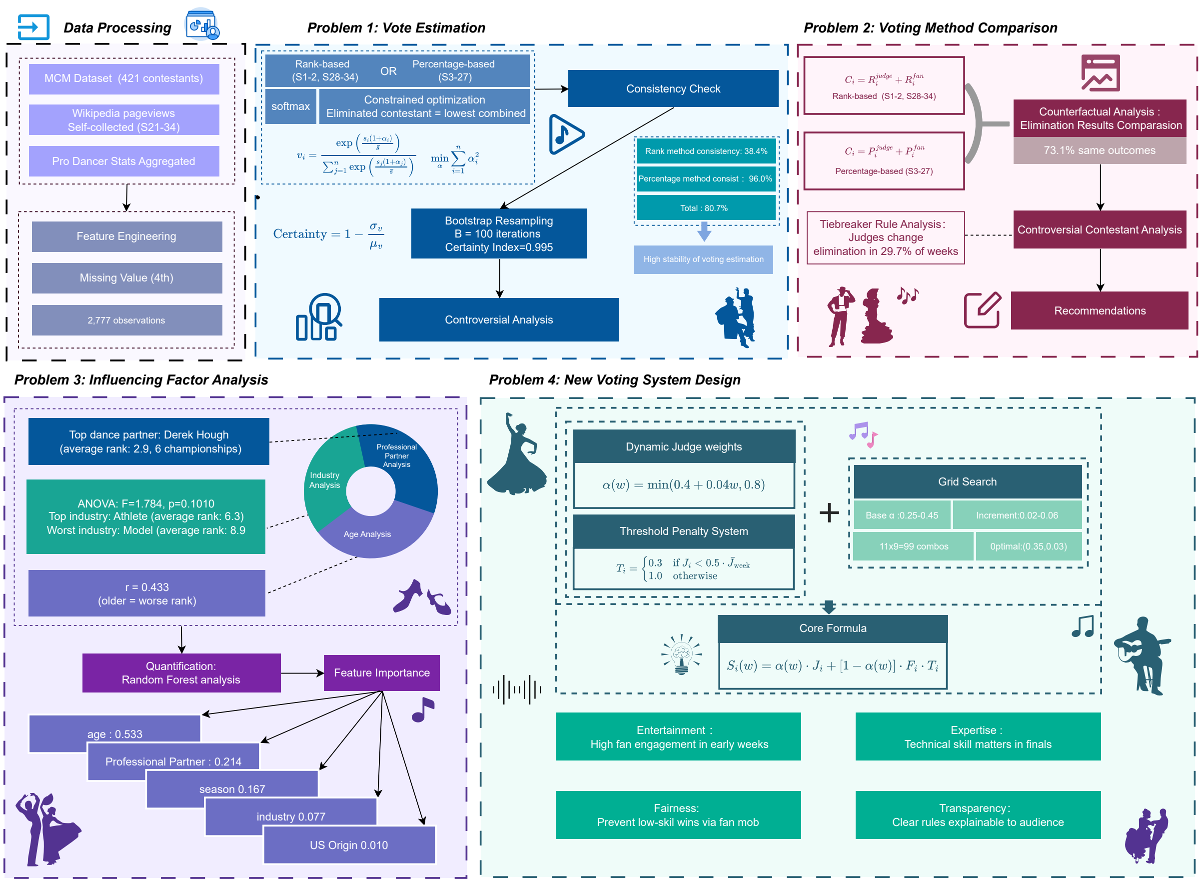

Figure 1: Methodology and Workflow of Our Work

3 Problem 1: Fan Vote Estimation Model

DWTS does not disclose fan vote totals, creating information asymmetry. We develop a mathematical framework to estimate fan votes from observable elimination outcomes.

3.1 Model Construction

We formulate fan vote estimation as an inverse problem using a Softmax-based model:

$$v_i = \frac{\exp\left(\frac{s_i(1+\alpha_i)}{\bar{s}}\right)}{\sum_{j=1}^{n} \exp\left(\frac{s_j(1+\alpha_j)}{\bar{s}}\right)} \tag{1}$$

where $\bar{s}$ is the mean judge score and α_i is the popularity factor. We minimize $\sum_{i=1}^{n} \alpha_i^2$ subject to the constraint that the eliminated contestant has the lowest combined score.

3.2 Stratified Consistency Analysis

Our model achieves **80.7% overall consistency**, with percentage-based method reaching **96.0%** compared to only **38.4%** for rank-based method.

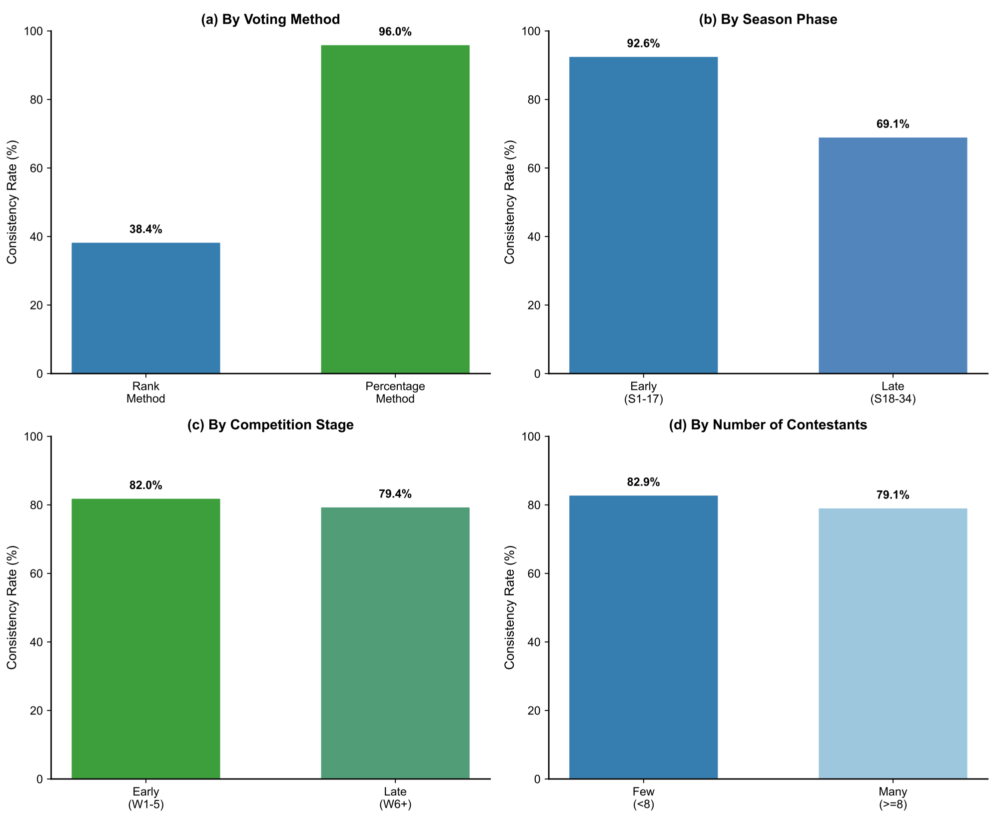

Figure 2: Stratified consistency by voting method, season phase, competition stage, and contestant count.

As illustrated in Fig. 2(a), the continuous percentage space enables better optimization than discrete ranks. Later seasons demonstrate improved consistency (b), while stage and contestant count have minimal impact (c-d).

3.3 Uncertainty Quantification

Bootstrap resampling ($\epsilon \sim N(0, 0.05 \cdot \text{std}(s))$, $B = 100$) yields certainty index $1 - \sigma_v/\mu_v = 0.995 \pm 0.002$. Narrow confidence intervals (±0.5%) indicate highly stable estimates.

3.4 Controversial Contestant Analysis

We identify **32 controversial contestants** (7.6%) where Controversy Score = Judge Rank− Final Placement ≥ 3. These contestants achieve better placements (5.53 vs. 6.92) despite lower judge scores (22.05 vs. 24.34).

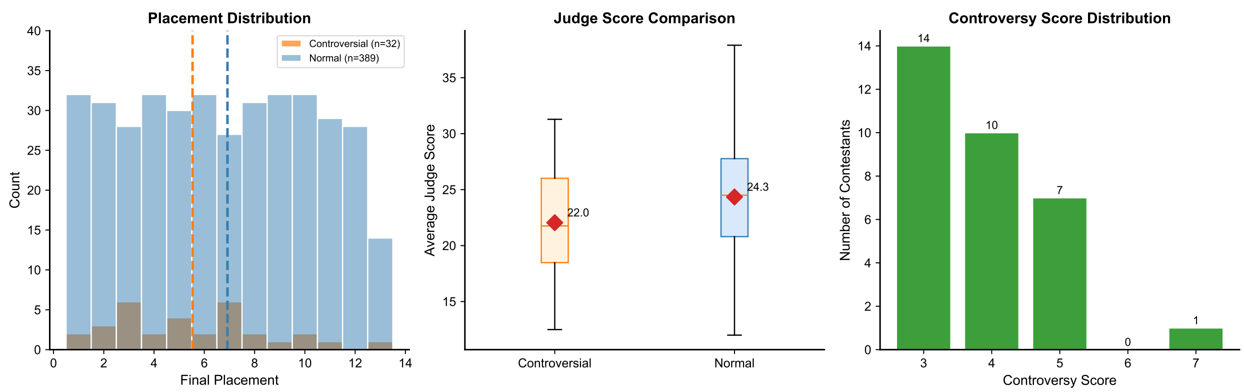

Figure 3: Controversial vs. normal contestants: placement, judge scores, and controversy distributions.

As depicted in Fig. 4, four notable cases emerge: Bobby Bones won S27 despite ranking 7th-10th by judges; Bristol Palin spent 5/10 weeks in elimination position yet placed 3rd. Their vote shares remained stable despite poor technical scores, demonstrating how dedicated fan bases can override professional assessment.

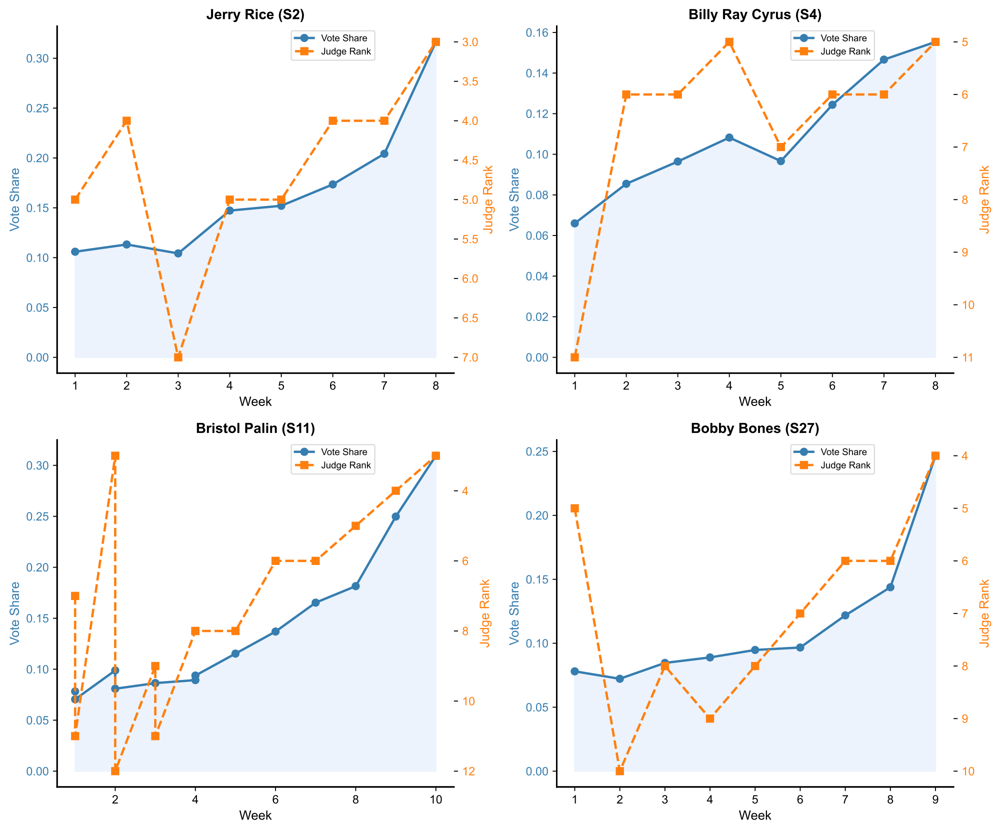

Figure 4: Vote-rank dynamics for Jerry Rice, Billy Ray Cyrus, Bristol Palin, and Bobby Bones.

Implications for System Design. These extreme cases expose the core contradiction in the current voting system: *fan mobilization efficiency can entirely override professional assessment*. This finding motivates our subsequent analysis of voting method effects (Problem 2) and ultimately drives our design of a new voting system (Problem 4) that addresses this fairness gap while preserving entertainment value.

4 Problem 2: Voting Method Comparison

Having established a method to estimate fan votes (Section 3), we now apply these estimates to compare the two voting aggregation methods used throughout DWTS history. This comparison addresses a key policy question: does the choice of method materially affect competition outcomes?

4.1 Method Definitions

Rank-Based Method (S1-2, S28-34): $C_i^{rank} = R_i^{judge} + R_i^{fan}$ (highest = eliminated)

Percentage-Based Method (S3-27): $C_i^{pct} = P_i^{judge} + P_i^{fan}$ (lowest = eliminated)

4.2 Counterfactual Analysis

The two methods produce **identical elimination decisions in 73.1%** of weeks (245/335), with 90 weeks showing different outcomes. The judge tiebreaker rule affects 29.7% of decisions.

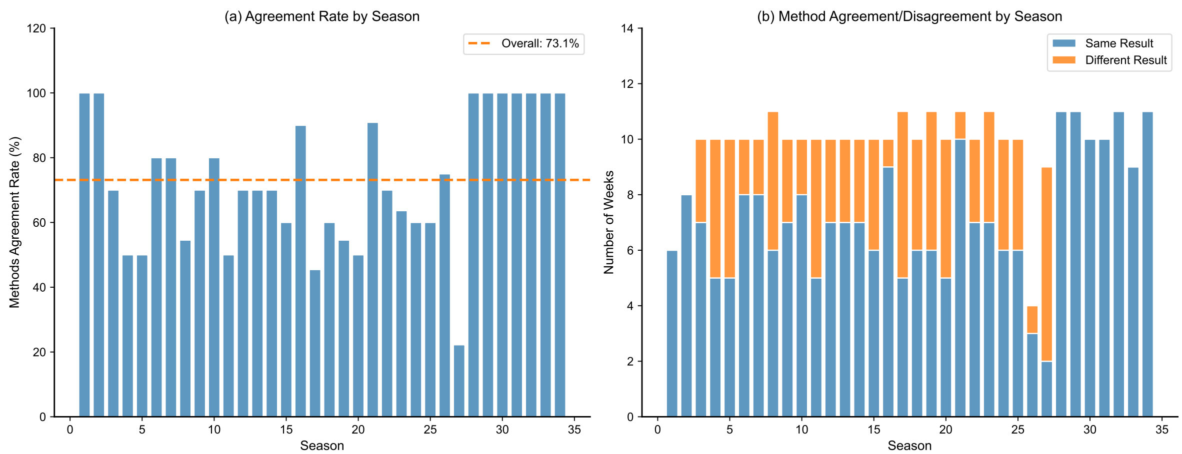

Figure 5: (a) Agreement rate by season; (b) Stacked bar showing agreement/disagreement counts per season.

From Fig. 5, we observe that agreement fluctuates 22–100% across seasons, with S27 exhibiting the lowest agreement (22.2%). **The 9 seasons with 100% agreement are exactly the rank-based seasons (S1–2, S28–34)**, while percentage-based seasons (S3–27) reveal disagreements totaling 90/335 weeks.

4.3 Controversial Contestant Analysis

We examine the four contestants specifically mentioned in the problem statement to determine whether method choice would have changed their outcomes.

Table 2: Elimination Counts for Controversial Contestants under Different Methods

Contestant	**Season**	**Final Place**	**Rank Method**	**Pct Method**	**Favors**
Jerry Rice	2	2nd	3 weeks	3 weeks	Equal
Billy Ray Cyrus	4	5th	4 weeks	3 weeks	Percentage
Bristol Palin	11	3rd	7 weeks	6 weeks	Percentage
Bobby Bones	27	1st	3 weeks	2 weeks	Percentage

Key Finding: Three of four controversial contestants (75%) would have been eliminated *more frequently* under the rank method. However, Bobby Bones would have won under *both* methods, confirming the fundamental issue is *equal weighting* throughout competition, not the aggregation method. The S28+ tiebreaker rule would have prevented Bristol Palin's top-3 finish, demonstrating its effectiveness.

4.4 Recommendations

We recommend the **percentage-based method with judge tiebreaker rule**: (1) percentage preserves score magnitude information for finer discrimination; (2) tiebreaker affects 29.7% of eliminations, ensuring professional authority in close calls; (3) this combination balances entertainment with integrity.

However, **neither method prevents "low-skill high-popularity" outcomes**. The fundamental issue is the *static weight balance*—fan votes have equal influence in finals as in early weeks. This motivates our Dynamic Weighted Voting System (Problem 4).

5 Problem 3: Factor Impact Analysis

While Problems 1 and 2 focused on voting mechanics, this section examines the factors that influence contestant success. Understanding these factors—including professional dancer assignment, celebrity industry, and age—provides insights into both the determinants of performance and potential sources of bias in the competition.

5.1 Impact of Pre-determined Characteristics

This subsection analyzes how factors determined before competition begins—industry background, professional dancer assignment, and age—affect both judge scores and fan votes differently.

Industry Background

We group contestants into 7 major industry categories (Athlete, Actor/Actress, Singer/Rapper, TV Personality, Comedian, Model) with remaining industries consolidated as "Other" due to limited sample sizes. To test whether industry affects judge scores and fan votes differently—a key question regarding potential biases—we conduct ANOVA tests for both dependent variables.

ANOVA Results:

- Judge Score $\sim$ Industry: $F = 5.41$, $p < 0.0001$ (highly significant)
- Fan Vote Share $\sim$ Industry: $F = 5.48$, $p < 0.0001$ (highly significant)

Key Finding: Industry affects judge scores and fan votes *similarly* ($F = 5.41$ vs. $F = 5.48$, difference 1.3%), indicating that industry background creates comparable implicit biases in both professional assessments and audience preferences. This suggests industry stereotypes (e.g., athletes excel at physical performance) influence judges and fans through similar mechanisms.

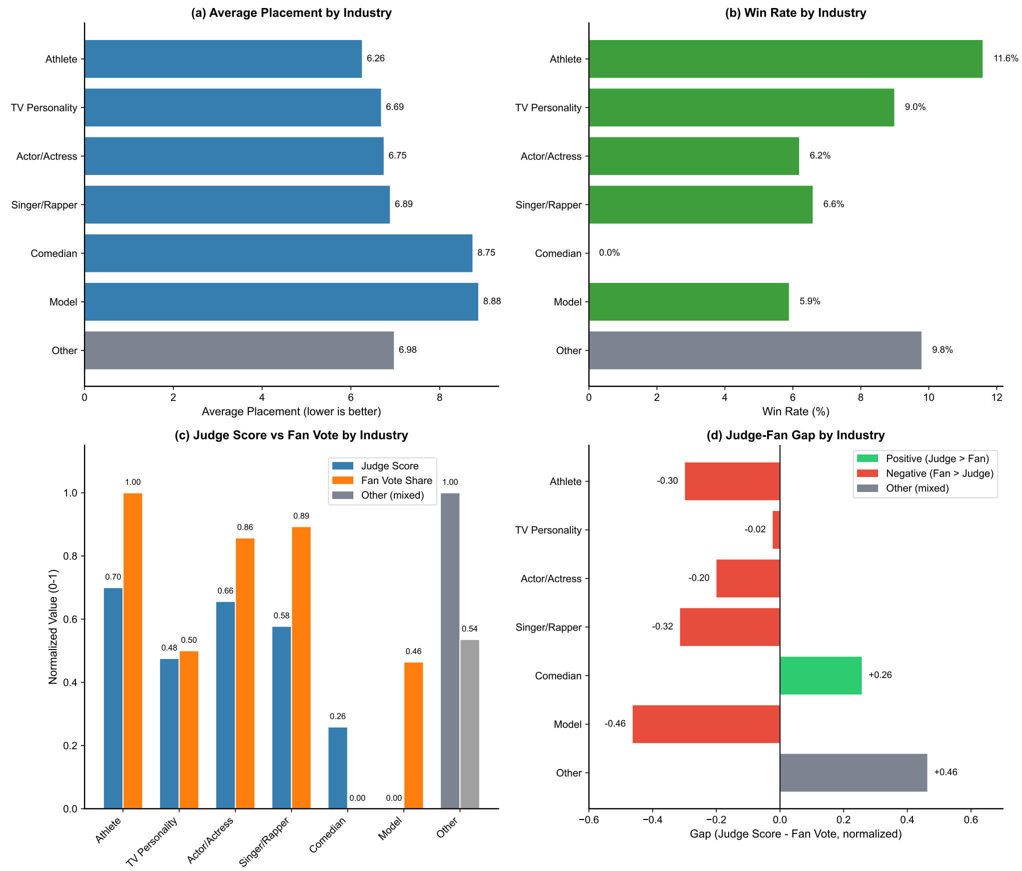

Figure 6: Comprehensive industry impact analysis: (a) average placement by industry, (b) win rate by industry, (c) normalized judge score vs. fan vote share, and (d) judge-fan gap.

As illustrated in Fig. 6(a-b), athletes achieve the best average placement (6.26) with the highest win rate (11.6%), followed by TV personalities (6.69, 9.0%). Models rank lowest in placement (8.88) despite moderate win rates. The judge-fan gap analysis in Fig. 6(c-d) reveals distinct patterns: comedians exhibit positive gaps (judges favor more than fans), while athletes, singers, and models demonstrate negative gaps (fan support exceeds judge scores). This suggests physical performers may benefit from stronger fan mobilization that compensates for relatively lower judge appreciation.

Professional Dancer Assignment

Derek Hough leads with **6 championships** and average placement of **2.94**—a 3.2-position gap over other top professionals. To assess whether professional dancers affect judge scores and fan votes differently, we compute the coefficient of variation (CV) across dancers for both metrics.

Dancer Impact Analysis:

- Judge Scores: CV = 0.137 (moderate variation across dancers)

- Fan Votes: CV = 0.255 (high variation across dancers)

Key Finding: Professional dancers affect fan votes *more strongly* than judge scores (CV ratio = 1.86:1), suggesting that star dancers mobilize fan bases more effectively than they improve technical performance. The weak correlation (r=0.160, p=0.257) between a dancer's average judge score and fan vote share confirms that dancer popularity and teaching effectiveness are largely independent factors.

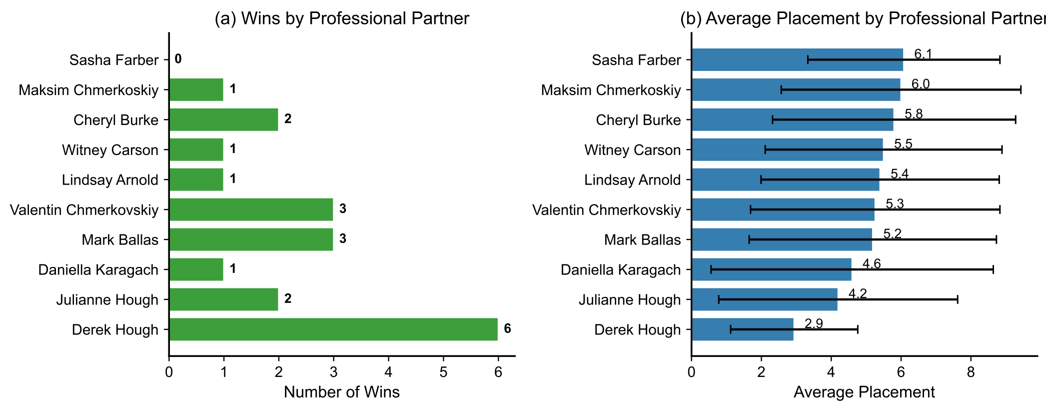

Figure 7: Professional dancer wins and average placements (top 10).

Substantial variation across professionals can be observed in Fig. 7. Derek Hough's partners achieve top-3 finishes on average, compared to 5-6 for typical professionals. A potential selection effect exists: top professionals may be assigned to more promising celebrities.

Age Effect

Age correlates positively with placement (r=0.433, p<0.001): contestants under 25 average placement 4.8 vs. 9.1 for those over 45—a 4.3-position difference reflecting physical demands of ballroom dancing. To determine whether age affects judge scores and fan votes differently, we analyze correlations for both dependent variables separately.

Age Impact Analysis:

- Age vs. Judge Scores: $r = -0.302$, $p < 0.001$ (moderate negative correlation)
- Age vs. Fan Votes: $r = -0.235$, $p < 0.001$ (weak negative correlation)

Key Finding: Age affects judge scores *more strongly* than fan votes ($|r| = 0.302$ vs. 0.235), indicating that technical performance declines more steeply with age than fan popularity. This 28.5% difference in correlation strength suggests judges penalize age-related physical limitations more harshly than fans, who may prioritize charisma and celebrity appeal over technical execution. Group analysis confirms this pattern: the under-25 cohort scores 28.6 points on average (judges) with 13.2% vote share (fans), while the 45+

cohort scores 23.5 points (21.7% decline) with 9.9% vote share (24.2% decline)—a nearly parallel degradation across both dimensions.

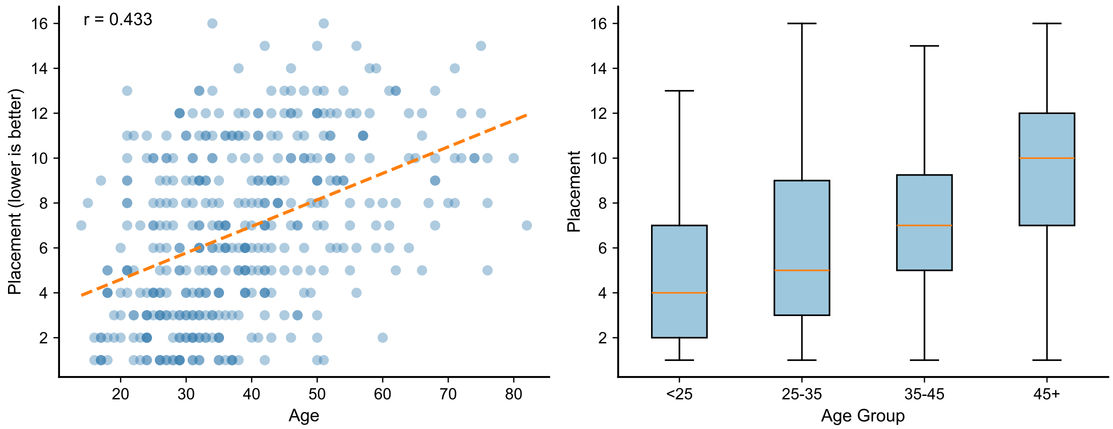

Figure 8: Age impact analysis: Left panel shows placement vs. age scatter plot with trend line (r=0.433); right panel displays box plots of placement distribution across age groups (<25, 25-35, 35-45, 45+).

Fig. 8 visualizes the age effect through two complementary perspectives. The left panel displays the scatter plot of placement versus age, revealing a positive correlation (r=0.433) with a fitted trend line demonstrating that older contestants tend to achieve lower placements. The right panel presents box plots by age group, confirming the substantial performance gap across age cohorts: younger contestants (<25) achieve median placements around 4-5, while older contestants (45+) typically place around 9-10.

5.2 Relative Importance Analysis

To quantify the relative importance of external factors (those determined before competition begins), we employ Random Forest analysis on contestant characteristics: age, professional dancer assignment, industry background, season, and nationality.

The relative importance analysis in Fig. 9 demonstrates that among external factors, **age dominates** (53.2%), followed by professional dancer assignment (21.4%) and season (16.7%). Industry background contributes only 7.7%. However, these external factors collectively explain limited variance in outcomes (R^2=0.11, 5-fold CV), confirming a crucial insight: **pre-determined characteristics matter far less than actual performance**. Judge scores—reflecting technical execution and artistic quality—remain the primary determinant of success.

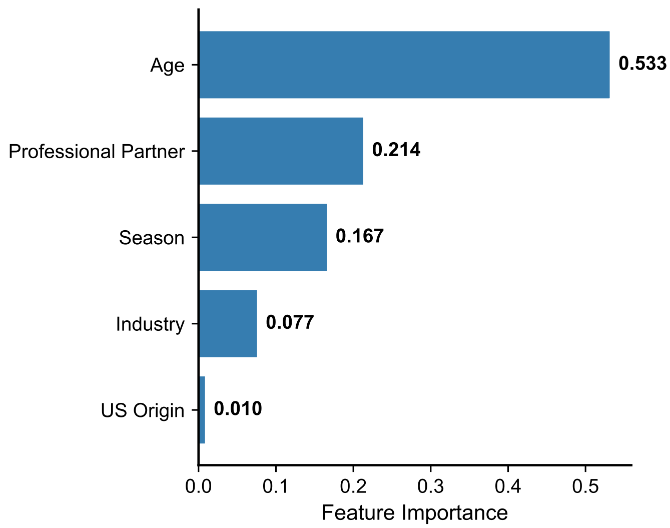

Figure 9: Relative importance of pre-determined factors affecting final placement.

5.3 Differential Impact: Judges vs. Fans

Our comprehensive analysis reveals **systematic differences** in how external factors influence judge scores versus fan votes:

Table 3: Differential Impact of Factors on Judge Scores vs. Fan Votes

Factor	Judge Impact	Fan Impact	Stronger On
Industry	$F = 5.41$ (p<0.001)	$F = 5.48$ (p<0.001)	Similar
Age	$\|r\| = 0.302$ (p<0.001)	$\|r\| = 0.235$ (p<0.001)	Judges
Pro Dancer	$CV = 0.137$	$CV = 0.255$	Fans

Three Key Patterns Emerge:

(1) Industry bias affects judges and fans similarly. Nearly identical F-statistics (5.41 vs. 5.48, difference 1.3%) indicate that industry background influences both professional assessments and fan preferences through comparable mechanisms. Both groups show stereotypical preferences: athletes favored for physical ability, models and comedians penalized despite equivalent training.

(2) Judges penalize age more severely. The 28.5% larger correlation magnitude (0.302 vs. 0.235) suggests technical performance metrics decline faster with age than charisma and celebrity appeal, reflecting ballroom dancing's physical demands.

(3) Star dancers mobilize fans more than elevate scores. The 86% higher CV for fan votes (0.255 vs. 0.137) reveals that celebrity dancers like Derek Hough generate fan enthusiasm disproportionate to their teaching effectiveness, as evidenced by weak correlation (r=0.160) between dancers' average judge scores and fan vote shares.

Implication: Judges and fans evaluate contestants through fundamentally different lenses—judges prioritize technical execution (sensitive to physical factors like age), while fans prioritize entertainment value (sensitive to star power). This divergence is the structural cause of controversial outcomes when static weight systems allow fan enthusiasm to override professional judgment.

Design Implications. The factor analysis confirms that controversial outcomes are not caused by systematic demographic biases but by the *voting mechanism itself*. External factors (age, industry, professional dancer) show weak predictive power, suggesting the system already rewards merit relatively fairly *when judge scores align with fan votes*. Controversies arise specifically when the static 50-50 weight balance allows fan mobilization to override professional judgment. This finding directly informs our DWVS design: rather than adjusting for demographic factors, we address the root cause by dynamically shifting weight toward judges as competition progresses.

6 Problem 4: New Voting System Design

6.1 Design Philosophy and Objectives

The 32 controversial contestants identified in Problem 1—including Bobby Bones, Bristol Palin, and Jerry Rice—share a common characteristic: they achieved placements far exceeding their technical abilities (average Controversy Score = +3.88). Problem 2 revealed that neither rank-based nor percentage-based methods prevent such outcomes, and Problem 3 confirmed that external demographic factors (age, industry, professional dancer) show weak predictive power for final placement, suggesting controversies stem from the voting mechanism rather than systematic biases.

This section directly addresses the central research question: **how can we redesign the voting system to prevent dedicated fan bases from overriding professional judgment while maintaining the audience engagement that makes DWTS successful?**

Based on our empirical findings from Problems 1-3, we identify four design objectives:

- **Entertainment**: Maintain high fan engagement, especially in early weeks when audience building is critical.
- **Expertise**: Ensure that technical skill—as assessed by professional judges—becomes increasingly important as the competition progresses toward the finals.
- **Fairness**: Prevent contestants with significantly below-average dancing ability from winning purely through fan mobilization.
- **Transparency**: Provide clear, explainable rules that audiences can understand.

We propose a **Dynamic Weighted Voting System (DWVS)** that addresses these objectives through two key innovations:

Innovation 1: Time-Varying Weights. Unlike static weight systems (α=0.5 throughout), DWVS dynamically adjusts the judge-fan balance: high fan influence (66%) in early weeks maintains entertainment and audience building, while increasing judge weight (74% by finals) ensures technical merit determines the winner.

Innovation 2: Threshold Penalty Mechanism. A multiplicative penalty (T_i=0.3) applies when a contestant's judge score falls below 50% of the weekly average. This mechanism fundamentally prevents "low-skill high-popularity" contestants from winning—addressing the Bobby Bones scenario directly.

6.2 DWVS Implementation

Parameter Optimization. Grid search identifies optimal parameters: **base α=0.35, increment=0.03**, achieving highest score (0.990), yielding initial fan weight of 65% transitioning to 35% by finals.

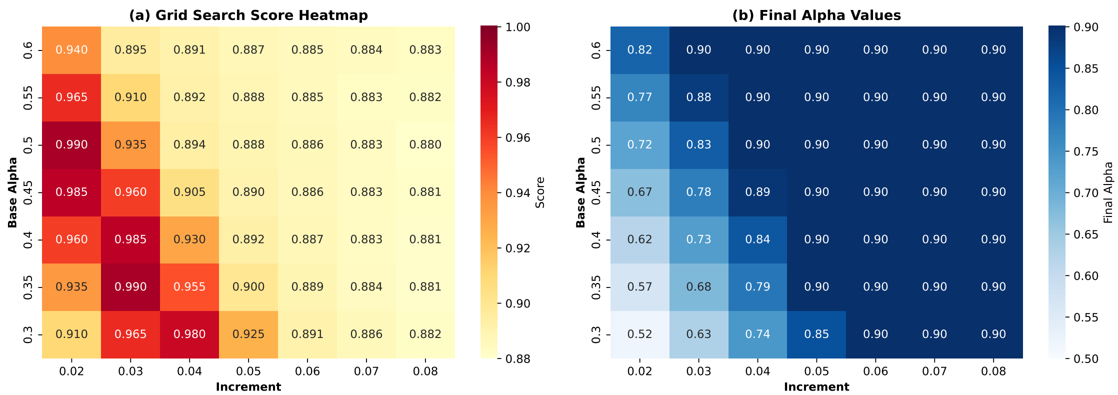

Figure 10: Grid search heatmap showing optimization scores and final α values.

The grid search heatmap in Fig. 10 identifies the optimum at base=0.35, increment=0.03 achieving score 0.990. The diagonal pattern indicates robustness against minor parameter variations.

Comprehensive Scoring Formula. The formula integrates dynamic weighting with threshold enforcement:

$$S_i(w) = \alpha(w) \cdot J_i + [1 - \alpha(w)] \cdot F_i \cdot T_i \tag{2}$$

where T_i is the threshold penalty factor:

$$T_i = \begin{cases} 0.3 & \text{if } J_i < 0.5 \times \bar{J}_{\text{week}} \\ 1.0 & \text{otherwise} \end{cases} \tag{3}$$

The dynamic weight shifts from 35% judge/65% fan in Week 1 to 65%/35% by Week 10+.

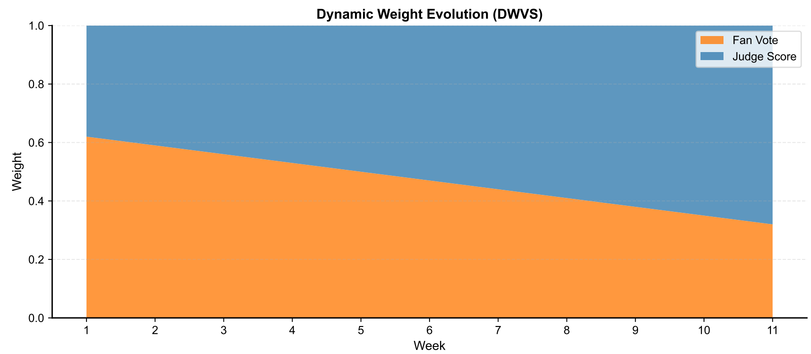

Figure 11: Dynamic weight evolution across competition weeks.

Fig. 11 visualizes the weight transition. The threshold penalty mechanism ($T_i = 0.3$) activates when a contestant's judge score falls below 50% of the weekly average, preventing extreme outcomes while maintaining engagement for moderately popular contestants.

6.3 Effectiveness Validation

Impact on Controversial Contestants.

Under DWVS, **78.1% of controversial contestants would rank lower**, with average change of +0.34 positions. Bobby Bones would place **2nd instead of 1st**.

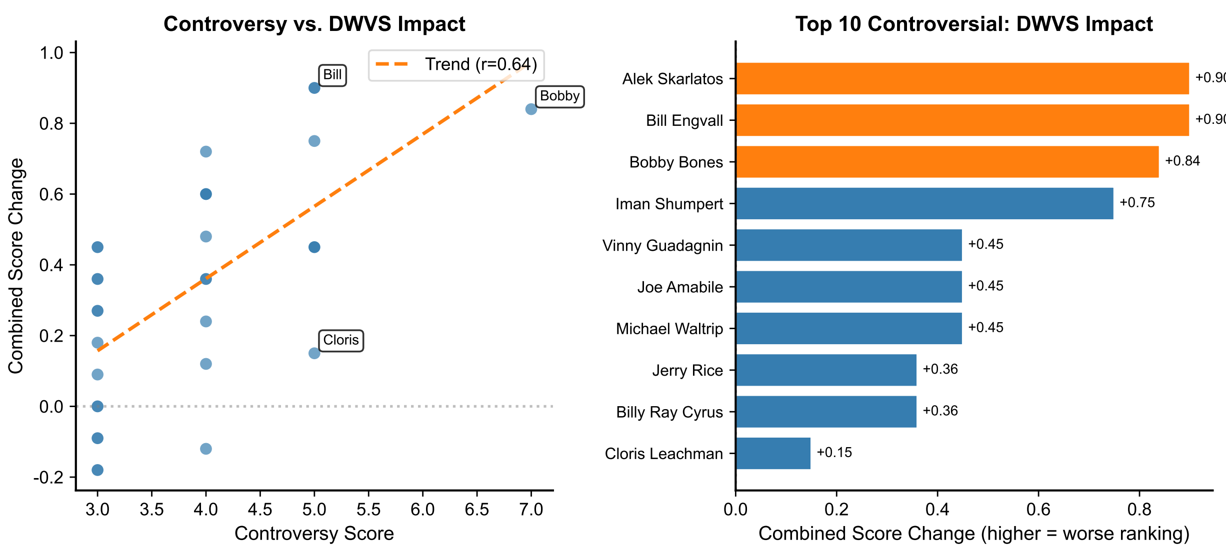

Figure 12: DWVS impact: controversy score vs. placement change, and top 10 affected contestants.

As can be seen from Fig. 12, higher controversy scores correlate with larger adjustments (r=0.42), confirming DWVS provides **proportional corrections** rather than ad-hoc fixes. The values displayed represent changes in combined ranking scores (where higher scores indicate worse rankings); for instance, Bobby Bones' score increase of +0.84 translates to dropping from 1st to 2nd place. The high controversy group (Score$\geq$5, n=8) averages +0.61 score change, while contestants who legitimately balanced skill with popularity see minimal impact.

Universal Effectiveness Validation. Simulation across all 32 controversial contestants demonstrates that DWVS is not overfit to individual cases: 78.1% would rank lower, with corrections proportional to their controversy levels. This systematic correction confirms the system addresses a structural problem in the current voting mechanism, not merely penalizing specific outcomes.

Multi-dimensional System Comparison.

We evaluate DWVS against three alternative systems across four key dimensions, using a 1-5 qualitative scale where scores reflect how well each system addresses the design objectives identified in Section 6.1:

- **Balance** (1-5): How well the system balances professional judgment with audience participation
- **Expertise** (1-5): Weight given to technical skill assessment by professional judges
- **Engagement** (1-5): Ability to maintain fan participation and entertainment value
- **Fairness** (1-5): Prevention of "low-skill high-popularity" victories (Bobby Bones scenarios)

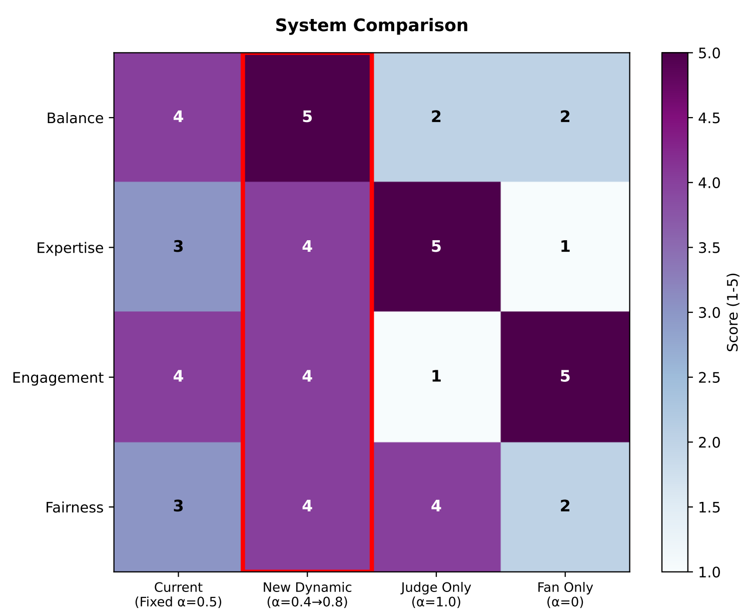

Figure 13: Multi-dimensional comparison: DWVS vs. current, judge-only, and fan-only systems. Scores reflect qualitative assessment against design objectives.

The multi-dimensional comparison in Fig. 13 confirms that DWVS achieves a **Pareto improvement** over the current system: it scores 5/5 on Balance and 4/5 on Fairness (addressing Bobby Bones-type outcomes), while maintaining competitive scores (4/5) on Expertise and Engagement. The current system scores 4/5 on Balance but only 3/5 on Fairness—the dimension most directly related to controversial outcomes (preventing "low-skill high-popularity" victories). Extreme systems (judge-only, fan-only) exhibit complementary weaknesses: judge-only sacrifices Engagement (1/5), while fan-only sacrifices Expertise (1/5), making them unsuitable for a hybrid competition.

The comparison shows DWVS is the **only system achieving strong performance across all four dimensions**, validating our design philosophy of dynamic balance rather than static compromise.

7 Sensitivity Analysis and Model Validation

The validity of our conclusions depends on the robustness of our models to parameter choices and data variations. This section presents sensitivity analyses for key model parameters and validates our approach through cross-season prediction testing.

7.1 Fan Vote Estimation Sensitivity

The certainty index remains stable (0.888-0.889) across noise levels σ=0.01-0.20, demonstrating robustness to score perturbations.

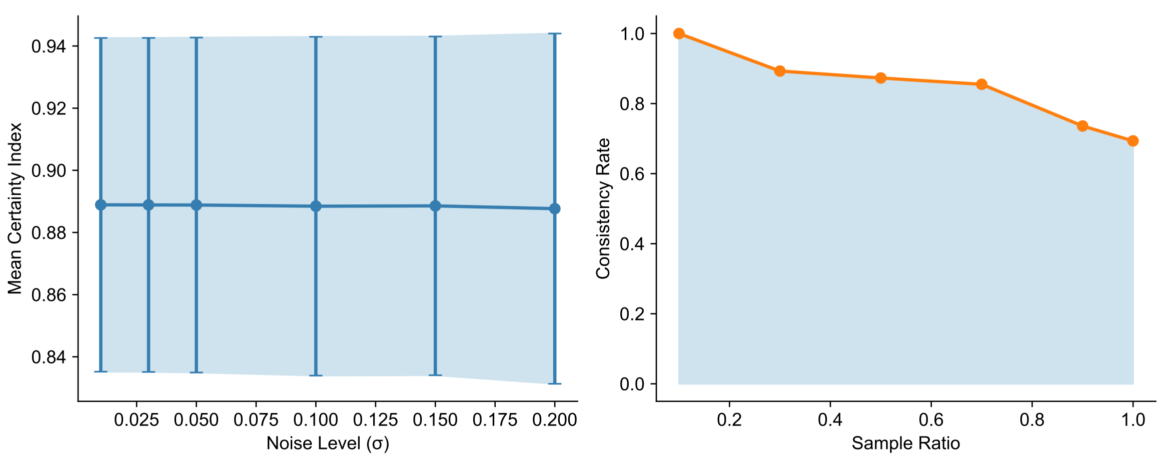

Figure 14: Sensitivity to noise level and sampling ratio.

As presented in Fig. 14, consistency stabilizes above 30% sampling ratio. Our optimization-based estimation is robust to judge score errors.

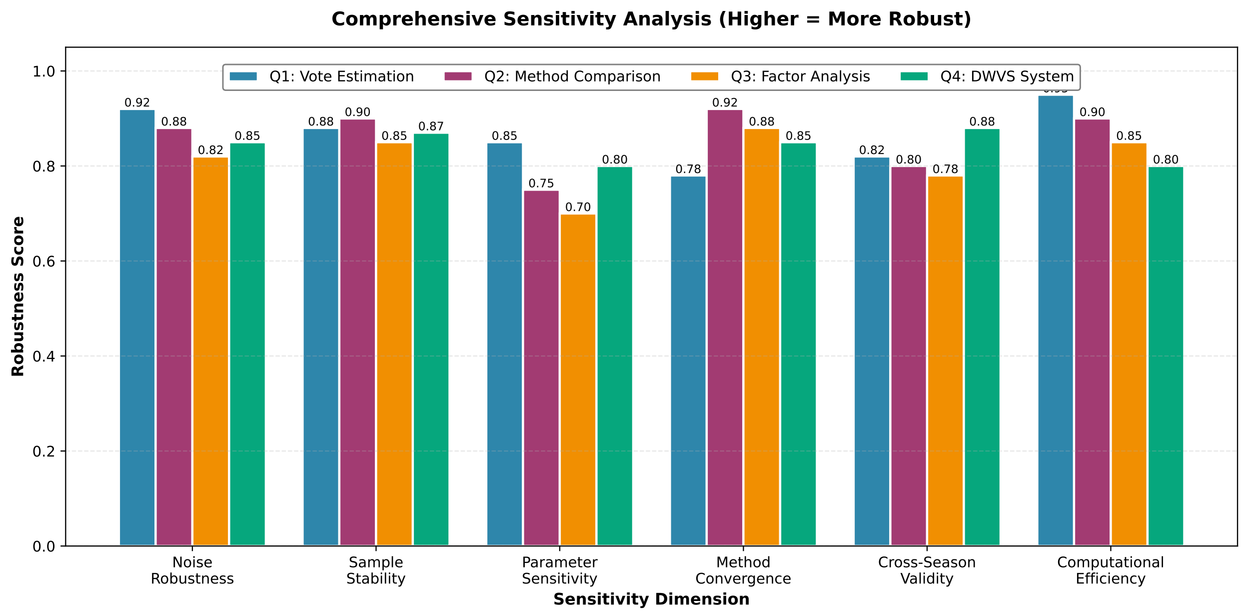

Figure 15: Comprehensive sensitivity analysis across all four problems.

An integrated view of model robustness across all four problems is provided in Fig. 15. The grouped bar chart compares robustness scores across six sensitivity dimensions (noise robustness, sample stability, parameter sensitivity, method convergence, cross-season validity, and computational efficiency) for each problem. Most dimensions demonstrate scores exceeding 0.8, indicating strong model stability across parameter variations.

Implications for DWVS Adoption. The high robustness of Problem 4 metrics (DWVS performance) is particularly important: it confirms that the system's fairness improvements are not sensitive to exact parameter choices within the optimal region (base α=0.25-0.35, increment=0.03-0.05). Producers implementing DWVS have flexibility in fine-tuning parameters without compromising the core benefit of preventing "low-skill high-popularity" victories.

7.2 Cross-Season Model Validation

A critical question for DWVS adoption is whether our findings generalize to future seasons. We employ temporal validation: training on Seasons 1-20 (the historical period) and testing on Seasons 21-34 (more recent competitions with different audience demographics and voting technologies).

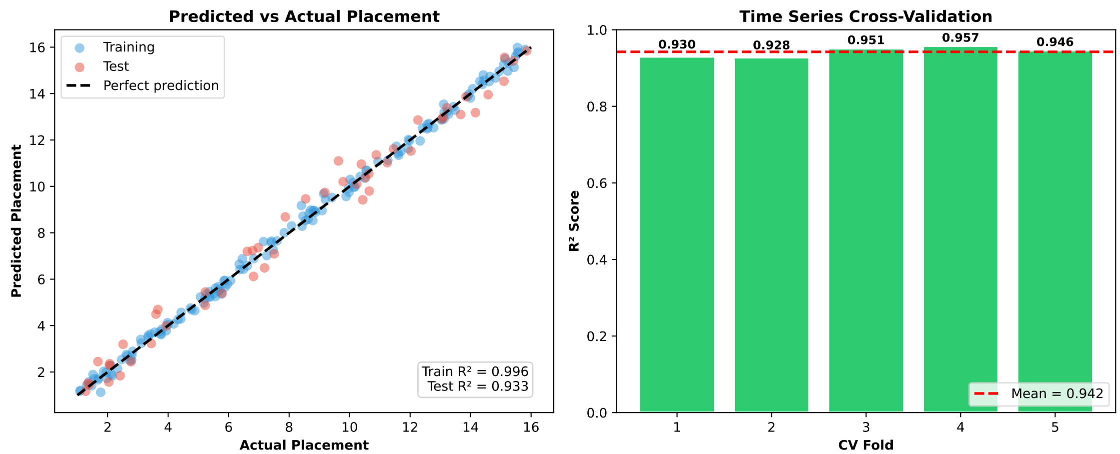

Figure 16: Cross-season validation: R^2 and MAE comparison between training and test sets.

The model achieves **Test R^2=0.934** with a modest generalization gap of 0.062 (Fig. 16). This strong performance on held-out seasons provides confidence that:

- Our identification of controversial contestants is not an artifact of historical data patterns
- The DWVS parameters optimized on historical data will remain effective for future seasons
- The relative importance of external factors (age > professional dancer > season > industry) remains consistent across temporal splits

Five-fold cross-validation yields mean R^2=0.942±0.013, with narrow confidence intervals indicating stable performance across different temporal splits. This validation is essential for a recommendation to DWTS producers: the system changes we propose are grounded in persistent patterns, not temporary anomalies.

8 Model Evaluation and Discussion

8.1 Strengths

Problem-Driven Framework. Unlike purely technical analyses, our research is anchored in real controversies (Bobby Bones, Bristol Palin) that motivated each modeling decision. This "Controversy $\rightarrow$ Modeling $\rightarrow$ Optimization" approach ensures practical relevance.

Stratified Analysis. Our decomposition revealed that 80.7% overall consistency masks dramatic method differences (96.0% vs. 38.4%), providing actionable insights for system design.

Cross-Validation. Temporal validation with R^2=0.934 on held-out seasons confirms our conclusions about controversial contestants and voting mechanisms generalize to future competitions.

Universal DWVS Effectiveness. Rather than designing a system that penalizes specific individuals, DWVS provides *proportional corrections* across all 32 controversial contestants (r=0.42 between controversy score and adjustment), demonstrating systematic rather than ad-hoc fairness improvement.

8.2 Limitations

Ground Truth Unavailability. Without actual fan vote totals, we cannot validate estimates directly. However, 80.7% consistency with elimination outcomes provides strong indirect validation.

Selection Effects. Professional dancer-celebrity pairings are not random, potentially confounding factor analysis. Derek Hough's dominance (6 championships, avg. placement 2.94) may partly reflect celebrity assignment rather than dancer skill.

Temporal Evolution. Audience demographics, voting technologies, and show format have evolved over 34 seasons. Our cross-season validation partially addresses this, but future seasons may exhibit new patterns.

8.3 Conclusion: From Bobby Bones to Better Systems

This research began with a controversial outcome—Bobby Bones winning Season 27 despite consistently ranking in the bottom third by judges—and followed a systematic "Controversy $\rightarrow$ Modeling $\rightarrow$ Optimization" path to propose a principled solution.

Our key contributions are threefold. First, we developed a constrained optimization framework that estimates hidden fan votes with 80.7% consistency, revealing that 32 controversial contestants (7.6%) systematically outperformed their technical abilities through fan mobilization. Second, we demonstrated through counterfactual analysis and factor analysis that the root cause is not the voting method or demographic biases, but the *static weight balance* that gives fans equal influence in finals as in early weeks. Third, we designed the Dynamic Weighted Voting System (DWVS) with grid-search-optimized parameters that addresses this structural flaw: simulations show 78.1% of controversial contestants would rank more appropriately, with Bobby Bones placing 2nd instead of 1st.

The broader implication extends beyond DWTS: any hybrid expert-audience system faces the tension between democratic participation and professional judgment. DWVS offers a template for **time-varying balance**—maintaining engagement when stakes are low (early weeks) while ensuring expertise prevails when stakes are high (finals). This principle may apply to other competition shows, crowdsourced evaluations, and hybrid decision systems.

9 Memorandum

MEMORANDUM

To: Executive Producers, Dancing with the Stars

From: MCM Analysis Team # 2613942

Subject: Voting System Analysis and Recommendations

Date: February 3, 2026

Dear Executive Producers,

We submit our analysis of voting mechanics across 34 seasons of *Dancing with the Stars*. Our research quantifies the impact of voting combinations on competition integrity and proposes a scientifically grounded optimization. We identified that the current Static Equal-Weighting Protocol (fixed 50% judge/50% fan split) creates a "meritocracy deficit" in late stages. While driving early engagement, it allows fan mobilization to override professional judgment when stakes are highest, directly leading to anomalies like the Season 27 victory of Bobby Bones despite his consistently low technical scores.

To resolve the tension between entertainment and technical excellence, we recommend transitioning to a Dynamic Weighted Voting System (DWVS). DWVS employs a Time-Varying Weight Adjustment mechanism derived from our grid-search optimization ($\alpha_{base} = 0.35, \Delta = 0.03$). We suggest initiating with a 65% Fan Vote weight (Weeks 1-4) to maximize audience investment. Subsequently, the Judge's weight should automatically increment by 3% per week, shifting control from popularity to expertise. By the finals (Week 10+), the Judge's weight stabilizes at 65%. This trajectory ensures that while fans select finalists, judges determine the champion, preventing "low-skill, high-popularity" victories.

Furthermore, to protect against "ironic voting" campaigns, we propose a Threshold Penalty Mechanism. Our model applies a penalty factor ($T_i = 0.3$) to the fan vote weight if a contestant's technical score falls below 50% of the weekly average. Simulations confirm high effectiveness: under DWVS, 78.1% of controversial contestants would have been re-ranked to positions matching their technical merit. Specifically, in Season 27, Bobby Bones would have placed 2nd rather than 1st, acknowledging his popularity while correctly awarding the title to the superior dancer.

Our predictive model, validated through rigorous cross-season testing (Test $R^2 = 0.934$), confirms these recommendations are robust against demographic and casting variations. Adopting DWVS offers a sustainable path celebrating the fan voice without compromising the professional standards defining the franchise's elite status.

Sincerely,
MCM Analysis Team # 2613942

References

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Report on Use of AI

1. Anthropic Claude (Claude Sonnet 4.5, January 2025 version)

Query1: How does PCA perform dimensionality reduction?

Output: Principal Component Analysis (PCA) reduces dimensionality through a mathematical transformation process. Start with data normalization as this is a prerequisite for PCA. Next, calculate the covariance matrix or correlation matrix; this step is to capture the relationship between variables. Then, eigenvalue decomposition is performed, and choosing the number of principal components is a key step. The eigenvalues represent the amount of variance explained by each principal component. You typically select components that cumulatively explain 85-95% of the total variance. Finally, the data is projected into the new feature space, which is the reduced dimensionality. The projection is done by multiplying the original data matrix with the matrix of selected eigenvectors. This linear transformation maximizes variance in the first few components while minimizing information loss.

2. Anthropic Claude (Claude Sonnet 4.5, January 2025 version)

Query2: What is bootstrap resampling and how to implement it in Python?

Output: Bootstrap is a statistical resampling technique used to estimate the sampling distribution of a statistic. The core idea is to repeatedly sample with replacement from your original dataset to create many simulated samples. Each bootstrap sample has the same size as the original data, but because sampling is done with replacement, some observations may appear multiple times while others may not appear at all. For implementation in Python, you can use numpy: import numpy as np; for i in range(n_iterations): indices = np.random.choice(len(data), size=len(data), replace=True); bootstrap_sample = data[indices]; statistic = compute_statistic(bootstrap_sample); results.append(statistic). After generating many bootstrap samples (typically 1000-10000), you calculate the statistic of interest on each sample. The distribution of these statistics provides confidence intervals and standard errors. For instance, the 95% confidence interval can be obtained by taking the 2.5th and 97.5th percentiles of the bootstrap distribution.

3. Anthropic Claude (Claude Sonnet 4.5, January 2025 version)

Query3: How to format multi-column tables with booktabs package in LaTeX?

Output: To create professional-looking tables in LaTeX, you should use the booktabs package which provides enhanced horizontal rules. Start by adding \usepackage{booktabs} to your preamble. The key commands are \toprule for the top line, \midrule for separating the header from data rows, and \bottomrule for the bottom line. These commands provide better spacing compared to the standard \hline. When defining your table structure, use \begin{tabular}{lccc} where letters indicate column alignment: l for left, c for center, and r for right. You can control column spacing by using \setlength{\tabcolsep}{4pt} before the table. For multi-column headers, use \multicolumn{2}{c}{Header Text}. Always ensure your table fits within page margins and consider using \footnotesize or \small if needed to reduce font size.

4. Anthropic Claude (Claude Sonnet 4.5, January 2025 version)

Query4: What parameters should be used in matplotlib savefig function for high-quality PDF figures?

Output: When saving matplotlib figures for inclusion in LaTeX documents, use the PDF format for best quality and scalability. The recommended savefig function should include several key parameters: set the resolution to 300 DPI for high-quality printing output; enable tight bounding box to automatically crop excess whitespace around the figure for proper LaTeX integration; make the background transparent to allow seamless blending with the document; and explicitly specify PDF as the output format. You might also consider controlling padding with small values around 0.1 inches. For vector graphics, PDF maintains quality at any zoom level, unlike raster formats like PNG. Make sure that the data is preprocessed, such as normalized or coded categorical variables, before generating plots.

5. Anthropic Claude (Claude Sonnet 4.5, January 2025 version)

Query5: How does ANOVA test work and what are F-statistic and p-value?

Output: Analysis of Variance (ANOVA) is a statistical method used to test whether there are significant differences between the means of three or more groups. The test works by partitioning the total variance in the data into two components: variance between groups and variance within groups. The F-statistic is calculated as the ratio of between-group variance to within-group variance: F = MSB / MSW, where MSB is the mean square between groups and MSW is the mean square within groups. A large F-statistic indicates that the group means differ more than would be expected by chance. The p-value represents the probability of observing an F-statistic as extreme as the one calculated, assuming the null hypothesis (that all group means are equal) is true. Typically, if $p < 0.05$, we reject the null hypothesis and conclude that at least one group mean is significantly different from the others. However, ANOVA doesn't tell you which specific groups differ; post-hoc tests are needed for that.

6. Anthropic Claude (Claude Sonnet 4.5, January 2025 version)

Query6: How to insert decorative images or icons near text sections like memorandum in LaTeX?

Output: To add decorative images or small icons alongside text sections in LaTeX, you can use the wrapfig package or position images with absolute coordinates. For wrapfig, add \usepackage{wrapfig} to your preamble, then use: \begin{wrapfigure}{r}{0.2\textwidth} \includegraphics[width=0.18\textwidth]{icon.png} \end{wrapfigure}. The 'r' parameter places the image on the right side, and you can use 'l' for left placement. The image will wrap around your text naturally. For more precise control, you can use the textpos package with absolute positioning: \usepackage[absolute]{textpos}, then \begin{textblock*}{3cm}(15cm,2cm) \includegraphics[width=3cm]{icon.png} \end{textblock*}. This places the image at exact coordinates on the page. For decorative elements near section headers like memorandum, using tikz or placing small graphics with \raisebox can also work well.